Raj Vaibhav

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Skills

Languages:	C++, C, JavaScript, Python, Java
Backend/Web:	Node.js, Express.js, REST APIs, HTTP, JSON, PostgreSQL
Systems/Tools:	POSIX Sockets, Multithreading, Linux, Git
Soft Skills:	Adaptable, Quick Learner, Problem-Solver

Projects

Webhook Delivery Service | GitHub — May' 26

- Built an **asynchronous webhook delivery service** in Node.js/Express that **decouples ingestion from delivery** - persisting incoming events and returning **202 Accepted** immediately while a background worker fans out deliveries out-of-band, so slow or failing subscribers never block ingestion.
- Designed a **PostgreSQL-backed job queue** using SELECT ... FOR UPDATE SKIP LOCKED for contention-free concurrent workers, with **at-least-once delivery**, **exponential backoff + jitter**, **idempotency keys**, and a **dead-letter** state after N failed attempts.
- Secured deliveries with **HMAC-SHA256** per-subscription payload signing and **API-key auth** on management endpoints.
- **Tech Stack:** JavaScript, Node.js, Express, PostgreSQL, REST, HMAC-SHA256

Mini HTTP Server | GitHub — Nov' 25

- Built a **multithreaded HTTP server** in C++ from scratch using **POSIX TCP sockets**, implementing the HTTP request/response lifecycle and **OS-level concurrency**.
- Designed and implemented a custom **thread pool** (using std::thread, mutex, and condition_variable) to handle **concurrent client connections** efficiently, supporting dynamic routing and static file serving.
- Benchmarked the server using **ApacheBench** (~10k+ req/sec locally) and analyzed latency, **concurrency scaling**, and CPU utilization to evaluate performance under load.
- **Tech Stack:** C++, POSIX Sockets, Thread Pool, HTTP, TCP/IP

Training

CSE Pathshala — Jun' 25 – Jul' 25

C++ Programming: OOPs and Data Structures

- Completed **35+ hours** of intensive live summer training focused on Object-Oriented Programming, Core C++, and DSA fundamentals.
- Practiced **implementation of key data structures** (arrays, linked lists, stacks, queues, trees, graphs) and **OOP principles** (classes, objects, inheritance, polymorphism, abstraction, encapsulation).
- Attained **problem-solving** proficiency by completing structured C++ assignments and passing a final assessment.

Certificates

- C++ Programming:OOPS and DSA | *CSE Pathshala* — Aug' 25
- Fundamentals of Network Communication | *Coursera* — Sept' 24
- Introduction to Hardware and Operating Systems | *Coursera* — Sept' 24

Achievements

- Accepted into **GirlScript Summer of Code (GSSoC) 2026;** selected to contribute to open source projects as part of a nationally recognized program.
- Solved 200+ algorithmic problems across **LeetCode**, **Hacker Rank** and **GeeksforGeeks** showcasing algorithmic thinking and strong data structures knowledge.

Education

Lovely Professional University — Phagwara, Punjab
Bachelor of Technology — Aug' 23 – Aug' 27
Computer Science and Engineering; CGPA: 7.32

Siddhartha Public School — Kanjhawala, Delhi
Intermediate — Mar' 21 – May' 22
PCM; Percentage: 73.6%

Rotary Public School — Gurugram, Haryana
Matriculation — Apr' 19 – Jan' 20
Percentage: 91%